



Air University
Final Examinations: Spring 2026

Student ID: _____

Subject: Database Systems
Class: BSCYS -VI
Section: B
Course Code: CYS-130

Time Allowed: 2Hrs
Max Marks: 50
FM's Name: Ms. Irum Aslam
FM's Signature: 

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.

CLO 2

Question 1

(15+15= 30 Marks)

A university is implementing a smart access control system to manage entry permissions across campus facilities. The system tracks students, their access cards, entry logs, emergency contacts, and programs.

Requirements:

1. Each student has a unique ID, full name (first name, middle name, last name), date of birth, and age (calculated from DOB).
2. Each student may have multiple phone numbers and one emergency contact. Emergency contacts include name and relationship to the student.
3. Each student is issued one access card. The card has a unique card number, issue date, and expiry date.
4. Each time a student enters a facility, an entry log is created with timestamp and location. Entry logs are linked to access cards.
5. Emergency contacts cannot exist independently in the system.
6. Each student is enrolled in exactly one program. Programs have a unique code and name.

a) Draw an ER diagram for the scenario above. Your diagram must clearly represent:

- Entities and attributes
- Composite attributes
- Multivalued attributes
- Derived attributes
- Key attributes
- Relationships and their cardinality ratios
- Any dependent entities and identifying relationships

b) Convert your ER diagram into a relational database schema. List all tables with their primary keys and foreign keys. Show how multivalued, composite, and derived attributes are handled.

CLO 4

Question 2

(2 + 2 + 2 + 4 + 5 + 5 = 20 Marks)

For all the following questions, you must use the Online Bookstore relational database schema given below:

- Book(BookID PK, Title, Author, Price, PublishedYear)
 - Customer(CustomerID PK, Name, Email)
 - Order(OrderID PK, OrderDate, CustomerID FK)
 - OrderDetails(OrderID FK, BookID FK, Quantity, PRIMARY KEY(OrderID, BookID))
- a) Write a query to list the titles of books priced higher than the average book price.
- b) Write a query to display customers who have placed more than 3 orders ordered by the number of orders descending.
- c) Write a query to list customers who have never placed an order.
- d) Create a view that shows each book's title and the total number of times it has been ordered.
- e) Identify the error in this query and explain what it will produce. Suggest the correct fix.
- ```
SELECT Author, Title, COUNT(*)
FROM Book
GROUP BY Author;
```
- f) Identify the error in this query and explain the consequence. Suggest the correct clause.
- ```
SELECT AVG(Quantity)
FROM OrderDetails
WHERE Quantity > AVG(Quantity);
```